Generate, Retrieve, or Calculate?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

TEACHER GUIDE · 1 OF 9

Generate, Retrieve, or Calculate?

> Prepare Generate, Retrieve, or Calculate?: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Distinguish retrieving a record, calculating from **input**s and generating wording

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Privacy by Choosing Less

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

Look for: Justify which information a stated task needs and remove unnecessary fields

Reference example, not the only acceptable wording: Keep topic choice; remove name, full date of birth, address and photograph. The totals do not require identity or contact details.

Collecting extra fields creates copies and exposure to manage even if those fields are later ignored or deleted.

Missing and Unbalanced Examples

Without opening the old lesson, explain one key idea from Missing and Unbalanced Examples. Give an example and a mistake that the idea helps you avoid.

Look for: Name the target collection, selection list and cases actually assessed

Reference example, not the only acceptable wording: The target is all 24 specified cards across four conditions. The sample is the 6 bright-circle cards. Dim circles, bright squares and dim squares are absent.

A frame containing only bright-circle IDs excludes every other condition, regardless of how randomly IDs are drawn from it.

Similarity

Without opening the old lesson, explain one key idea from Similarity. Give an example and a mistake that the idea helps you avoid.

Look for: **Calculate** similarity using explicitly selected features and a stated scoring rule

Reference example, not the only acceptable wording: A: 1, 1, 0 gives 2. B: 0, 0, 1 gives 1. C: 1, 0, 0 gives 1. A is the only highest scoring candidate among A, B and C under equal weights.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

> Three tasks are: invent a school motto, find the exact fire-drill meeting point in a supplied handbook, and divide 84 stickers among 7 teams. Choose **generation**, **retrieval** or **calculation** for each. Which **output**s need checking?

Concept to explain

Different systems create, look up, or compute answers in different ways.

Term	Meaning
generate	create a new output from learned patterns
retrieve	find and return stored information
calculate	apply defined operations to values
provenance	where an output or claim came from

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Keep operations distinct without presenting three mutually exclusive product categories. A modern system can **retrieve** records, call a calculator and **generate** a final response.

Lewis and colleagues provide a primary example combining **retrieval** and **generation**: **Retrieval**-Augmented **Generation** for Knowledge-Intensive NLP Tasks (2020), https://arxiv.org/abs/2005.11401.

Use this as teacher background, not required pupil reading. The fictional notices and arithmetic here are independent classroom examples.

Readiness answer

The task needs the three totals, not pupil identities. Use the least information needed and follow the teacher's approved tool and data rules.

The fictional N1 and N2 notice details printed in the student lesson

Paper for three-column **trace**s labelled source facts, **calculation** and wording

The independent library record B in W4; no online account is needed

End goal for the pictorial case

By the end, I can separate **retrieval**, **calculation** and **generate**d wording.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Generate, Retrieve, or Calculate? using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT question and available sources or rules

2. Process — the deciding step

PROCESS generate patterns, **retrieve** records, or **calculate**

3. Output — the result

OUTPUT answer with **mechanism**-appropriate checking

Key idea

CORE IDEA Different systems create, look up, or compute answers in different ways.

How the mechanism works

UNDER THE HOOD **Generation** synthesizes **output**s from learned distributions, **retrieval** selects stored records, and **calculation** applies formal operations.

Products often combine all three; verification should match the component that produced each claim.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE A story generator creates new wording, a library search **retrieve**s a stored title, and a calculator computes a result from stated operations.

Step 1

STEP 1 — OBSERVE question and available sources or rules

Step 2

STEP 2 — **TRACE generate** patterns, **retrieve** records, or **calculate**

Step 3

STEP 3 — CONCLUDE answer with **mechanism**-appropriate checking

Model answer

MODEL REASONING Check **generate**d claims against independent **evidence**, check **retrieve**d facts against the cited record and freshness, and recompute **calculation**s or inspect the operation. Mixed **output**s need more than one check.

A claim to avoid

DO NOT OVERCLAIM Fluent wording does not reveal whether a fact was **retrieve**d, **generate**d, or **calculate**d.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

A learner needs a new poem, 18+7, and a passage from the class handbook.

The teaching case

Choose the operation for each.

Exact explanation to model

Why the result follows

Generate the poem, **calculate** 25, **retrieve** the actual passage. Inventing a fluent passage is not **retrieval**.

Explain the added worked example

Explain the added worked example · 1

Three tasks can produce similar-looking text while doing different work. **Retrieval** finds existing information. **Calculation** applies mathematical operations to **input** values.

Generation produces content using a model and the context it receives. A single application can combine all three, so the label AI does not tell you which step happened.

Explain the added worked example · 2

Our fictional makers club has a current notice N2: Wednesday, Room 4, six tables with four places at each table, and three places reserved for helpers. An older notice N1 says Room 2. N2 explicitly replaces N1 for this event.

These records are invented for this lesson; no search, account or real booking is needed.

Explain the added worked example · 3

To answer “Which room does the current notice list?”, find N2 and return Room 4 with the record identifier. That is **retrieval**. To answer “How many places remain after the helper reservation?”, use 6 × 4 − 3 = 21.

That is **calculation**. To suggest a friendly invitation, **generate** wording and check that its factual details agree with N2.

Explain the added worked example · 4

Retrieving N1 accurately would still give an outdated room for this task. Successful **retrieval** means finding a record, not proving that every record is correct or current. Check the event, source and version.

Here the explicit replacement statement tells us which notice governs the classroom example.

A calculator given 6 × 4 correctly returns 24, but 24 is the total capacity, not the number remaining after three reserved places. Correct arithmetic on an incomplete expression answers the wrong question.

Explain the **input** values, units and operations before accepting a number.

Key terms

Term	Meaning
Retrieval	Finding and returning existing information from a collection
Calculation	Applying specified mathematical operations to input values
Generation	Producing content using a model and its input context

Term	Meaning
Source record	An existing item whose content and origin can be inspected
Trace	A record of the inputs, operations and outputs used in a task

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Draft statement	Evidence in this example	Decision
Meet Wednesday in Room 4	Current notice N2	Supported by the record
There are 21 non-helper places	6 × 4 − 3 from N2	Supported by calculation
Free snacks will be provided	Not stated in N2	Remove or verify separately

Model the reasoning aloud · 5

A **generate**d invitation might add free snacks because that detail fits a familiar invitation pattern. Plausible wording is not **evidence** that snacks are provided.

Remove that claim or obtain an appropriate confirming source before including it. **Generate**d content can be useful and can contain correct facts; its fluent style alone does not verify those facts.

Model the reasoning aloud · 6

Generation also does not guarantee originality. A response may repeat wording already present in its context or elsewhere.

Retrieval and **calculation** can feed a generator, and a generator can propose a **calculation** for another tool. Keep the **evidence** path visible instead of assuming that every sentence came from the same operation.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Generate, Retrieve, or Calculate?

A fictional craft invitation combines three different operations.

The facts we will use

> **Supplied facts**
>
> Approved record C: Tuesday, Hall 4, three tables with four learner places each.

> **Supplied facts**
>
> Two places are reserved for helpers; no snack information is supplied.

AI ACADEMY / AI-1/25

Generate, Retrieve, or Calculate?

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Retrieve facts

Read Tuesday, Hall 4 and 3 x 4 places from approved C.



2. Calculate

12 total places minus 2 reserved =10 learner places.



3. Generate wording

Draft a friendly invitation using only those supported details.



4. Check claims

A sentence promising snacks has no supporting record.

CHANGE ONE THING

Reserve four places instead of two, keeping the room and day unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Retrieve facts	Read Tuesday, Hall 4 and 3×4 places from approved C.
2. Calculate	12 total places minus 2 reserved =10 learner places.
3. Generate wording	Draft a friendly invitation using only those supported details.
4. Check claims	A sentence promising snacks has no supporting record.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Retrieval copies relevant source information; **calculation** follows arithmetic; **generation** proposes wording. These jobs can cooperate, but one does not validate the others.

A friendly **generate**d sentence can still contain an invented detail. Keep the approved source beside the draft and check each factual claim separately.

What this example does not establish

Keep the stated limit

The fictional source must itself be current and authoritative; **retrieval** alone does not ensure a record is correct.

Change one thing in the pictured case

Change one thing

Change: Reserve four places instead of two, keeping the room and day unchanged.

Predicted result

Prediction: Eight learner places remain.

Why the result follows

Reason: 3×4−4=8; this changes capacity for learners, not the **retrieve**d date or location.

Ask while showing the picture

Ask while showing the picture · 7

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Generate, Retrieve, or Calculate?, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A learner needs a new poem, 18+7, and a passage from the class handbook.

2. State the decision to explain

Write the question to explain: Choose the operation for each.

3. Reveal the worked reasoning

Reveal after **prediction**s: **Generate** the poem, **calculate** 25, **retrieve** the actual passage. Inventing a fluent passage is not **retrieval**.

Hold back the independent task

Keep for an independent attempt: Choose a method for each task: invent a dragon story; find the library's posted hours; compute 7 plus 5. Explain why the methods differ.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or **calculation** that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, **trace** conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Correct Thursday to Wednesday using N2; Room 4 is supported. Remove free snacks unless independently verified. Replace 24 remaining with 21 from 6 × 4 – 3.

One sentence: The makers club meets Wednesday in Room 4, with 21 places after the helper reservation. Exit: 6 × 4 – 5 = 19; day, room, table count and places per table stay the same.

The changed reservation invalidates the old remaining count.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve the minimal-input principle from Week 24.
7 to 20 minutes	Compare the three operations using N2 and complete W1.
20 to 33 minutes	Trace the diagram and complete W2. Ask what the result’s unit and meaning are.
33 to 45 minutes	Complete W3 and separate a retrieved record from a current record.
Session 2 start 0 to 8 minutes	Retrieve why correct arithmetic can still answer the wrong question.

Stage	Teacher action
8 to 23 minutes	Complete the guided correction. Require evidence for each factual claim.
23 to 38 minutes	Collect W4 independently, with source, expression, result and unsupported detail.
38 to 45 minutes	Complete W5 and the exit. Discuss identical outputs with different or unknown methods.

Reduce help deliberately

Completion practice

Completion task: A calculator correctly returns 24 for 6 × 4. A pupil reports this as the number of non-helper places available. Explain the problem and write the correct expression, result and meaning.

Reasoning to listen for

Teacher check: 24 is total capacity before the three helper places are reserved. The requested remaining count is 6 × 4 − 3 = 21 non-helper places.

The calculator can execute the supplied expression correctly while that expression omits a required step.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full **trace** only when the partial **trace** is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Generation retrieval** and **calculation**. Proposed explanation: “Retrieving a record proves it is current”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Check event and version; N1 is superseded by N2 in this example.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Label each arrow with its operation. Can the system add snacks because invitations often mention them?

Reasoned teacher answer

The arrows are **retrieval**, **calculation** and **generation**. No: snacks need confirmation; familiar wording is not **evidence**.

Changed-case reasoning: Eight learner places remain. 3×4−4=8; this changes capacity for learners, not the **retrieve**d date or location.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Generate, Retrieve, or Calculate? investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Classify nine **output**s by **mechanism** and verification need.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

> **Find the unsupported claim**
>
> CLAIM TO REPAIR Generating plausible text is the same as retrieving a fact.

For guessing, return to given facts and request a **trace**. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the **calculation**.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Retrieving a record proves it is current: Check event and version; N1 is superseded by N2 in this example.

A correct calculator result always answers the task: The expression must include the required reservation and correct units.

Fluent **generate**d details are verified facts: Free snacks and transport need **evidence**; wording alone is insufficient.

The final sentence reveals the method: Matching **output**s do not establish matching operations; inspect records and **trace**s.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Generate, Retrieve, or Calculate? questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Generate the poem, **calculate** 25, **retrieve** the actual passage. Inventing a fluent passage is not **retrieval**.

Independent question

Independent question · withhold the key

Choose a method for each task: invent a dragon story; find the library's posted hours; compute 7 plus 5. Explain why the methods differ.

Expected independent answer

Teacher key · Expected independent answer

Generate the story, **retrieve** the posted hours, and **calculate** 12. Creativity, source lookup and arithmetic require different **evidence** and checking.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Generating plausible text is the same as retrieving a fact.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Generate a motto and inspect suitability; **retrieve** and cite the actual handbook passage for the meeting point; **calculate** 84/7=12 stickers each and check 12×7=84. All need task-appropriate checks.

A fluent invented location cannot substitute for the handbook.

Explanations for the additional workbook examples

W1 Name the operation and its evidence

Workbook question · W1

For each task name the main operation and a suitable check: find the room listed in N2; work out six tables of four places minus three reserved places; suggest an inviting title for the event.

Can an application combine these operations?

Teacher reasoning · W1

Room: **retrieval**, checked against the right source N2 and event. Remaining places: **calculation**, checked as 6 × 4 − 3 = 21 with correct **input**s. Title: **generation**, checked for the requested style and unsupported factual details.

An application can combine them; the labels describe operations rather than exclusive types of application.

W2 Separate a correct sum from a correct answer

Workbook question · W2

A calculator correctly returns 24 for 6 × 4. A pupil reports this as the number of non-helper places available. Explain the problem and write the correct expression, result and meaning.

Teacher reasoning · W2

24 is total capacity before the three helper places are reserved. The requested remaining count is 6 × 4 − 3 = 21 non-helper places.

The calculator can execute the supplied expression correctly while that expression omits a required step.

W3 Inspect a record and a draft

Workbook question · W3

N1 lists Room 2. N2 explicitly replaces N1 for this event and lists Wednesday in Room 4. A draft says Thursday in Room 2 with free snacks. Identify the supported corrections and the unsupported detail.

Explain why finding N1 successfully would not settle the current-room question.

Teacher reasoning · W3

Use Wednesday and Room 4 from N2. Free snacks are unsupported and should be removed or verified with a suitable source.

N1 is an existing record but is superseded for this event, so retrieving it does not establish the current room.

W4 Trace a new library workshop

Workbook question · W4

Current fictional record B lists a library workshop on Friday in Room 5: five tables with six places each, with four places reserved for helpers. B replaces old record A, which lists Room 3.

A draft says “Thursday in Room 3, with 30 places remaining and a free bus home.” **Retrieve** the current day and room, **calculate** the non-helper places, write a corrected factual sentence and identify the detail that cannot be supported by B.

Teacher reasoning · W4

Retrieve Friday and Room 5 from B. **Calculate** 5 × 6 − 4 = 26 non-helper places. A suitable sentence is: The workshop is Friday in Room 5 with 26 places after the helper reservation, according to B and the capacity **calculation**.

The free bus is unsupported and must be removed or verified separately. The draft’s 30 is total capacity before reservation.

W5 Do not infer a method from its output

Workbook question · W5

Two applications both display “26 places remain.” Only one provides record B and a **trace** showing 5 × 6 − 4. What can and cannot be concluded from their matching sentences?

Describe how you would check the numerical claim without relying on either application’s assurance.

Teacher reasoning · W5

The visible numerical **output**s match, but that alone does not reveal the second application's method or prove that either checked the right task. Inspect current B, confirm the **input**s and independently **calculate** 5 × 6 − 4 = 26.

The first **trace** supports inspection; a verbal assurance of checking is not equivalent to inspectable **evidence**.

Assess independent transfer

Independent question · withhold the key

Selected case: Current fictional record B lists a library workshop on Friday in Room 5: five tables with six places each, with four places reserved for helpers. B replaces old record A, which lists Room 3.

A draft says "Thursday in Room 3, with 30 places remaining and a free bus home." **Retrieve** the current day and room, **calculate** the non-helper places, write a corrected factual sentence and identify the detail that cannot be supported by B.

Selected case answer

Teacher key · Selected case answer

Retrieve Friday and Room 5 from B. **Calculate** 5 × 6 − 4 = 26 non-helper places. A suitable sentence is: The workshop is Friday in Room 5 with 26 places after the helper reservation, according to B and the capacity **calculation**.

The free bus is unsupported and must be removed or verified separately. The draft's 30 is total capacity before reservation.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible **trace**.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Generate, Retrieve, or Calculate? to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

Label every sorter component as **generate**, **retrieve**, **calculate**, fixed rule, learned **prediction**, or **human decision**.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. **Retrieve** this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 26: Prompts Are Design Instructions. **Retrieve** today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use three coloured cards for recorded fact, derived quantity and suggested wording. Work with counters for total and reserved places.

Extension: explain which steps should be repeated if only the room changes versus if the number of tables changes.

W4 requires Friday, Room 5, record B, $5 \times 6 - 4 = 26$, a consistent corrected sentence and rejection of the unsupported free bus.

After support change B to reserve seven helper places: require 23 remaining with the same day and room, and continued rejection of the transport claim.

A numerical result without its source and expression is incomplete **evidence**.

Week 26 turns a vague request into a clear prompt with supplied **evidence**, **output** requirements and tests.

Plan the delayed return

Suggested return points in this level: ai-1/26, ai-1/28, ai-1/31. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Use the sorter's approved catalogue for facts, explicit **calculation**s for scores and checked prose for explanations.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Generate, Retrieve, or Calculate? lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Distinguish retrieving a record, calculating from **input**s and generating wording

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: A calculator correctly returns 24 for 6 × 4. A pupil reports this as the number of non-helper places available.

Explain the problem and write the correct expression, result and meaning.

Supported practice answer

Reasoned practice answer: 24 is total capacity before the three helper places are reserved. The requested remaining count is 6 × 4 − 3 = 21 non-helper places.

The calculator can execute the supplied expression correctly while that expression omits a required step.

Check every learner before discussion: Use the worked case for **Generation retrieval** and **calculation**. Proposed explanation: “Retrieving a record proves it is current”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Check event and version; N1 is superseded by N2 in this example.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Reserve four places instead of two, keeping the room and day unchanged. Require a **prediction** and a reason before comparing. Expected result: Eight learner places remain.

Reason: 3×4−4=8; this changes capacity for learners, not the **retrieve**d date or location.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

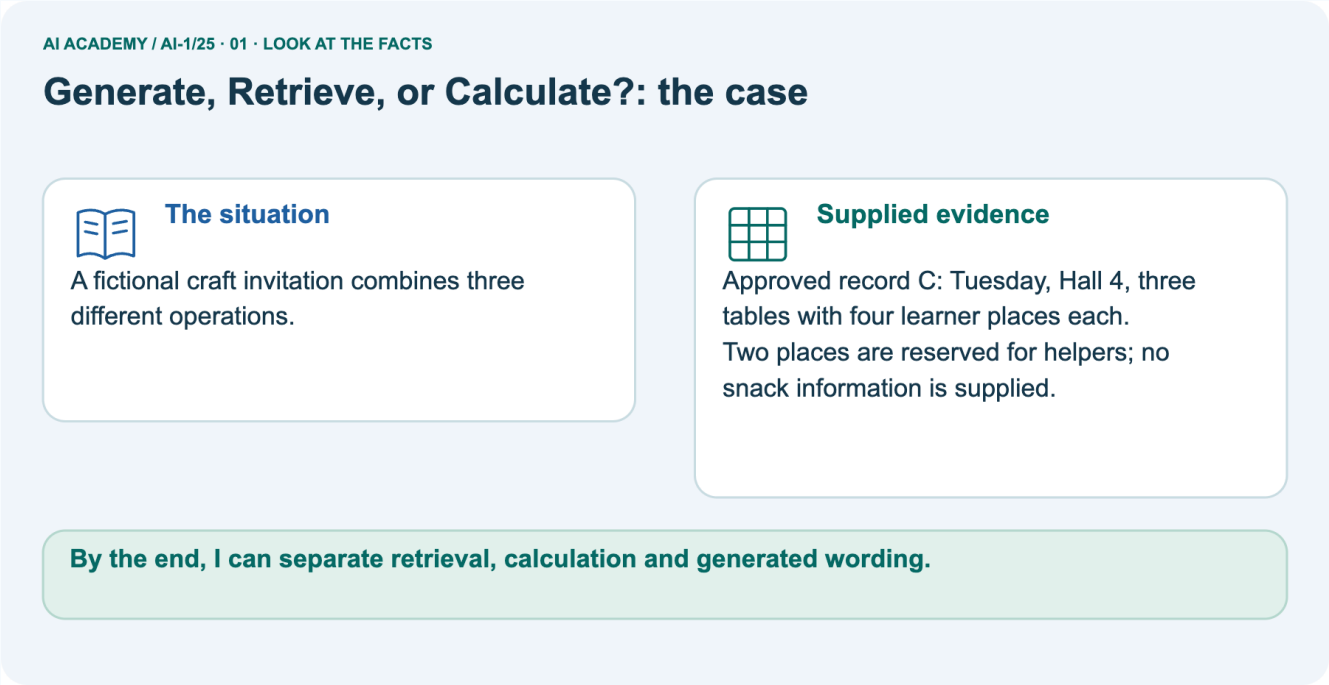
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, **trace**/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Generate, Retrieve, or Calculate? graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Generate, Retrieve, or Calculate?: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional craft invitation combines three different operations. Supplied facts: Approved record C: Tuesday, Hall 4, three tables with four learner places each.

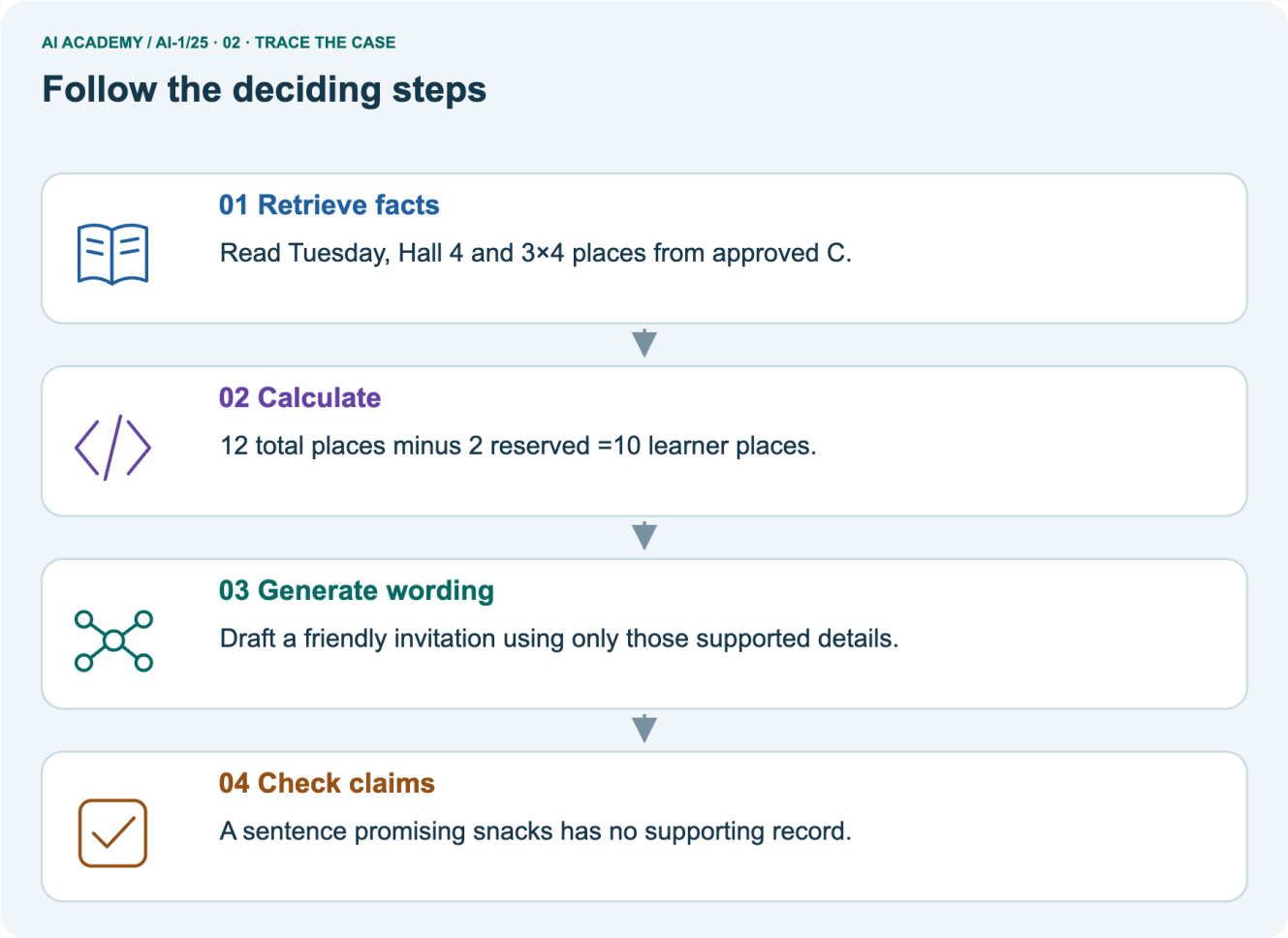
Two places are reserved for helpers; no snack information is supplied.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

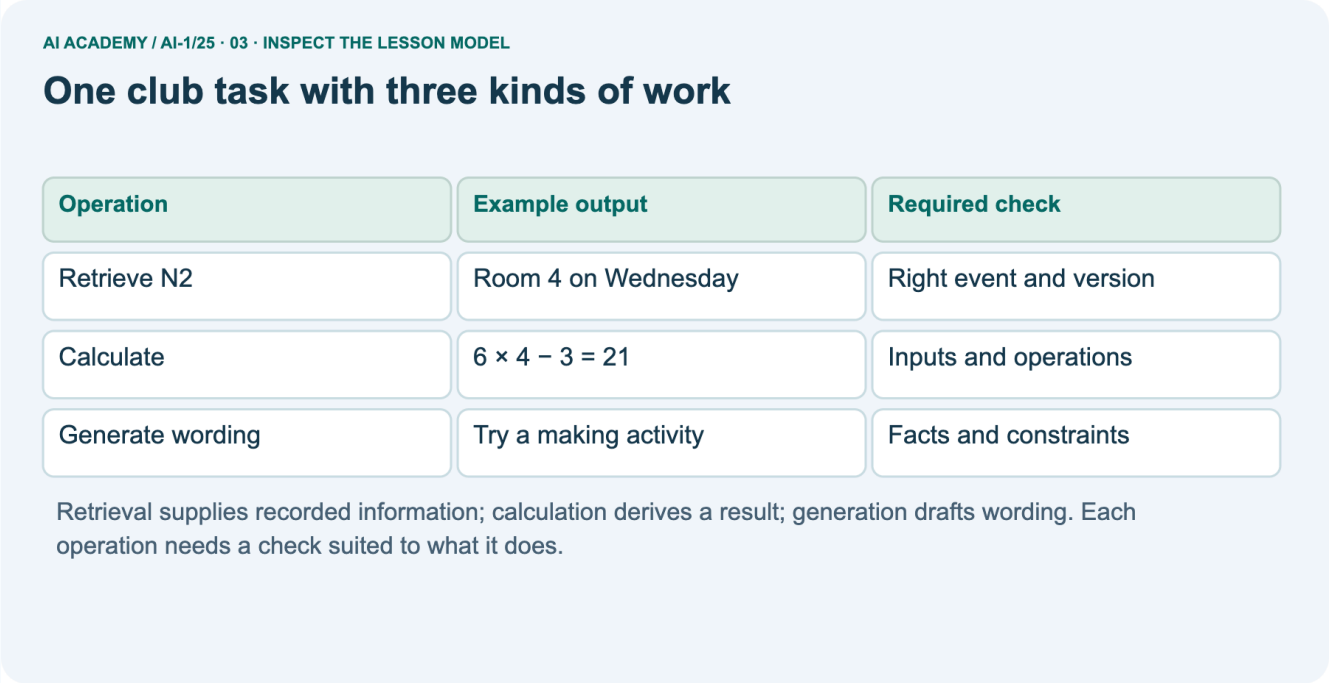
Retrieve facts: Read Tuesday, Hall 4 and 3×4 places from approved C. → **Calculate**: 12 total places minus 2 reserved =10 learner places. → **Generate** wording: Draft a friendly invitation using only those supported details. → Check claims: A sentence promising snacks has no supporting record.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. One club task with three kinds of work



Follow the labelled picture with the matching explanation. Open the vector diagram

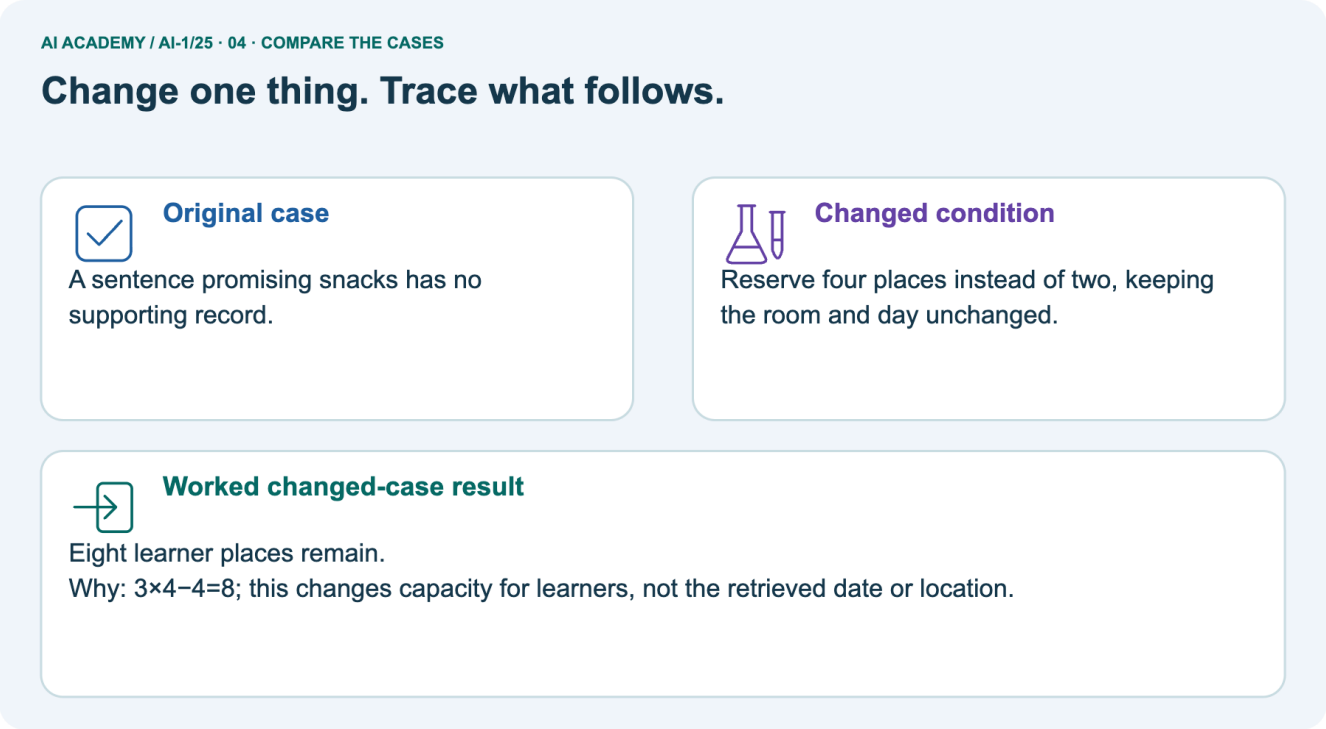
One club task with three kinds of work. **Retrieve** N2 | Room 4 on Wednesday | Right event and version. **Calculate** | 6 × 4 − 3 = 21 | **Input**s and operations. **Generate** wording | Try a making activity | Facts and constraints

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A sentence promising snacks has no supporting record. Changed condition: Reserve four places instead of two, keeping the room and day unchanged. Expected result: Eight learner places remain.

Reason: 3×4−4=8; this changes capacity for learners, not the **retrieve**d date or location.

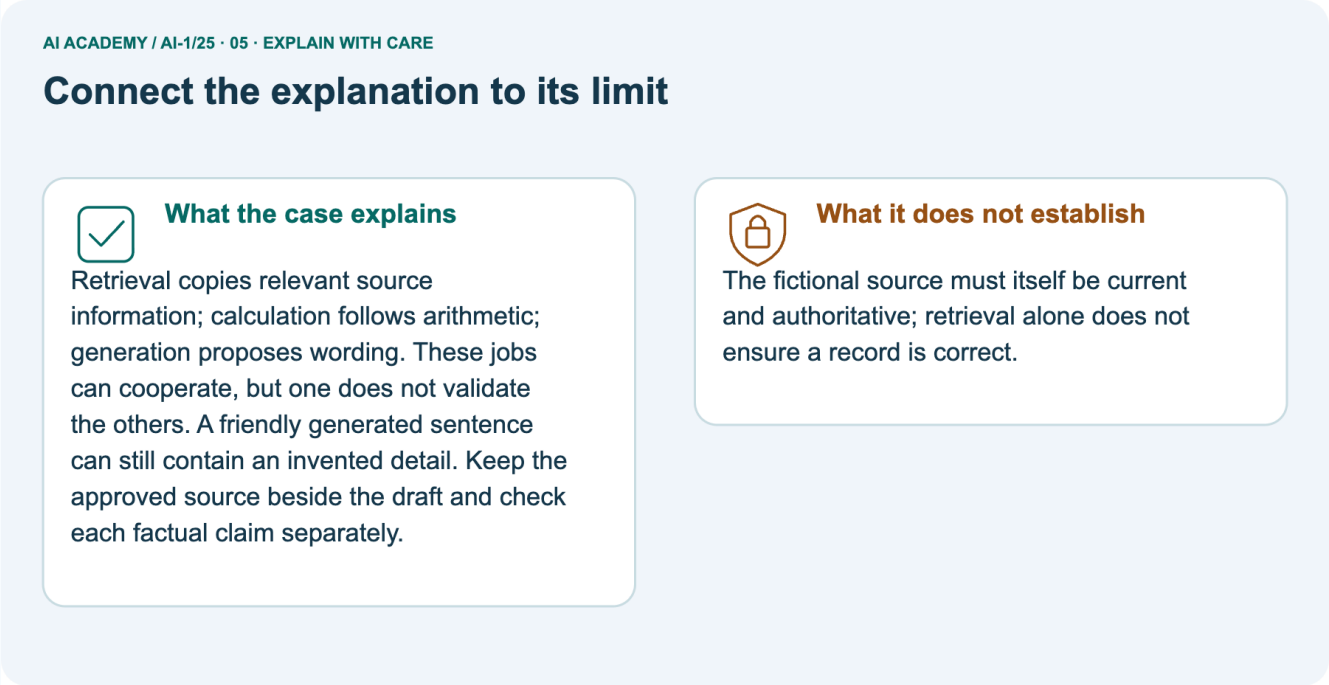
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: 3×4−4=8; this changes capacity for learners, not the **retrieve**d date or location.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieval copies relevant source information; **calculation** follows arithmetic; **generation** proposes wording. These jobs can cooperate, but one does not validate the others.

A friendly **generate**d sentence can still contain an invented detail. Keep the approved source beside the draft and check each factual claim separately.

Limit: The fictional source must itself be current and authoritative; **retrieval** alone does not ensure a record is correct.

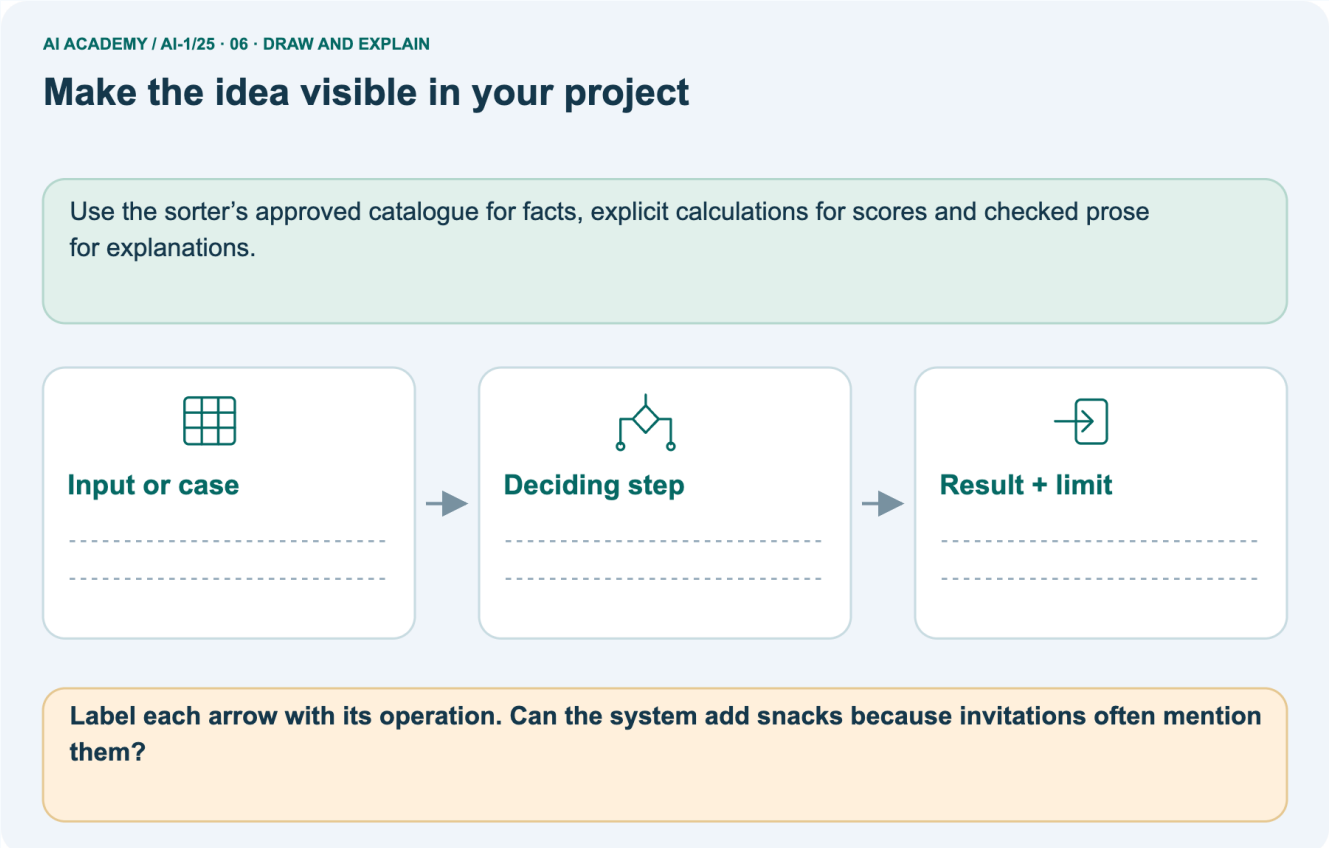
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: The fictional source must itself be current and authoritative; **retrieval** alone does not ensure a record is correct.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use the sorter’s approved catalogue for facts, explicit **calculation**s for scores and checked prose for explanations. Draw your **input**, deciding step, result and limit. Supported question: Label each arrow with its operation.

Can the system add snacks because invitations often mention them?

> Draw your project **trace** and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project **trace** and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The arrows are **retrieval**, **calculation** and **generation**. No: snacks need confirmation; familiar wording is not **evidence**.